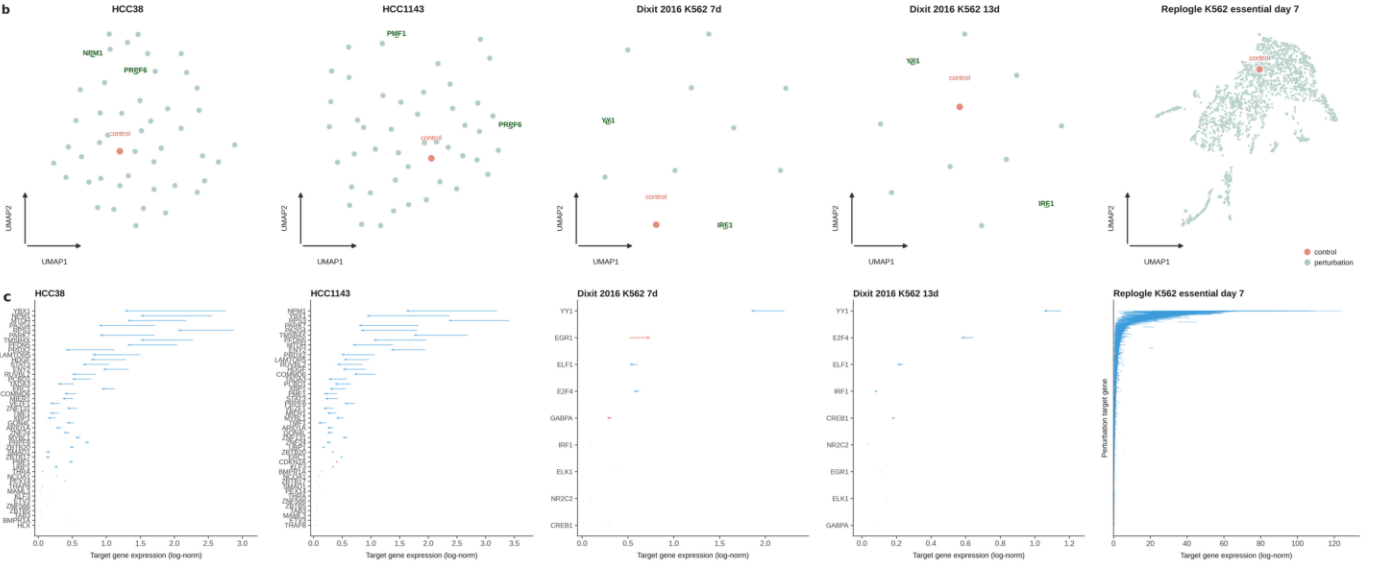


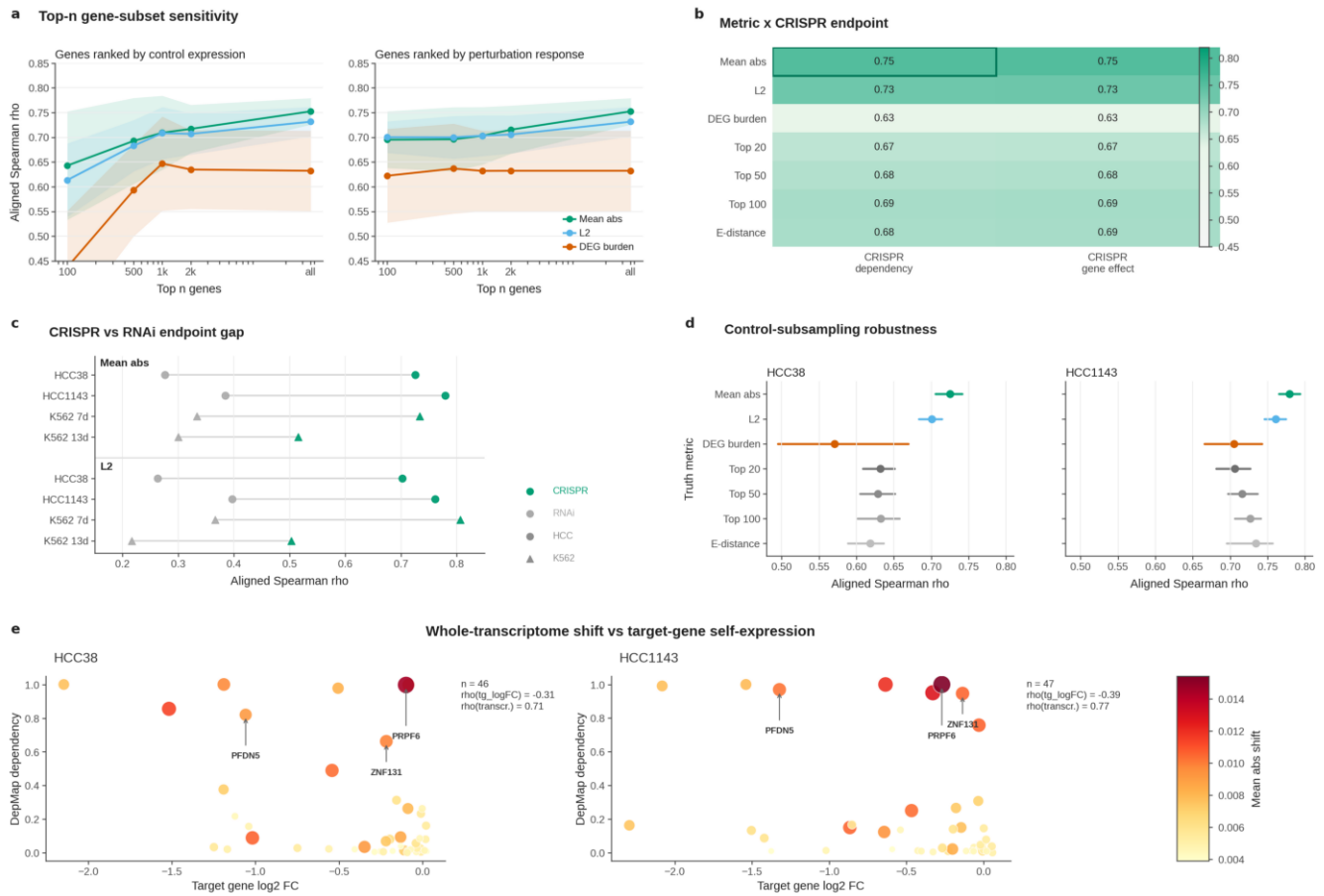
a

Dataset	Size	Benchmark use
Perturbation datasets		
HCC38	36,601 genes x 14,175 cells	47 targets 1,666 controls
HCC1143	36,601 genes x 11,405 cells	47 targets 1,325 controls
Dixit 2016 K562 7d	20,824 genes x 28,034 cells	10 targets 5,381 controls
Dixit 2016 K562 13d	20,824 genes x 15,849 cells	10 targets 3,491 controls
Replogle K562 essential day 7	8,563 genes x 310,385 cells	2057 targets 10,691 controls
Endpoint datasets		
DepMap CRISPR dependency (primary endpoint)	1,186 cell lines x 18,435 genes	HCC38 47 / HCC1143 48 / K562 7d 10 / K562 13d 10 / Replogle K562 1,882
DEMETER2 RNAi (sensitivity endpoint)	701 mapped cell lines x 17,309 genes	HCC38 48 / HCC1143 48 / K562 7d 9 / K562 13d 9 / Replogle K562 1,882



Extended Data Fig. 1 | Dataset familiarization and endpoint inputs

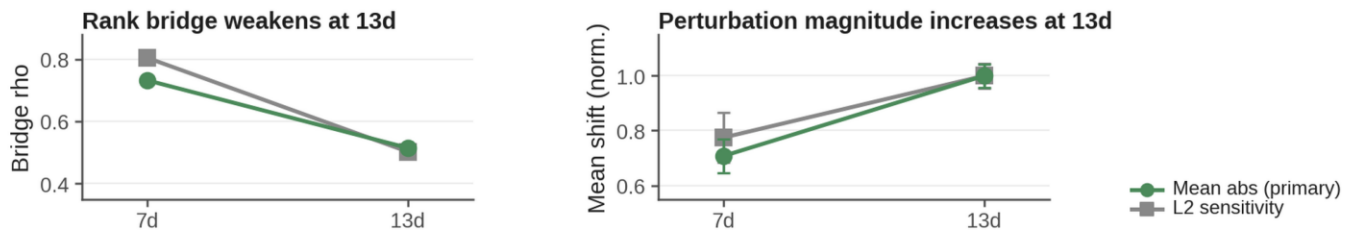
a, Dataset overview table. Five perturbation-expression contexts (HCC38, HCC1143, Dixit 2016 K562 7d, Dixit 2016 K562 13d, and Replogle K562 essential day 7 CRISPRi) and two endpoint datasets (DepMap CRISPR dependency and DEMETER2 RNAi) are summarized with cell-line identity, size, and benchmark use. b, UMAP of perturbation-level mean profiles for HCC38, HCC1143, Dixit 2016 K562 7d, Dixit 2016 K562 13d, and Replogle K562 essential day 7 (five panels, left to right). c, Target-gene expression change for the same five contexts (left to right). Each arrow represents one perturbation target: the base of the arrow indicates the target gene's mean expression in control cells, and the tip indicates its mean expression after perturbation (log-normalized).



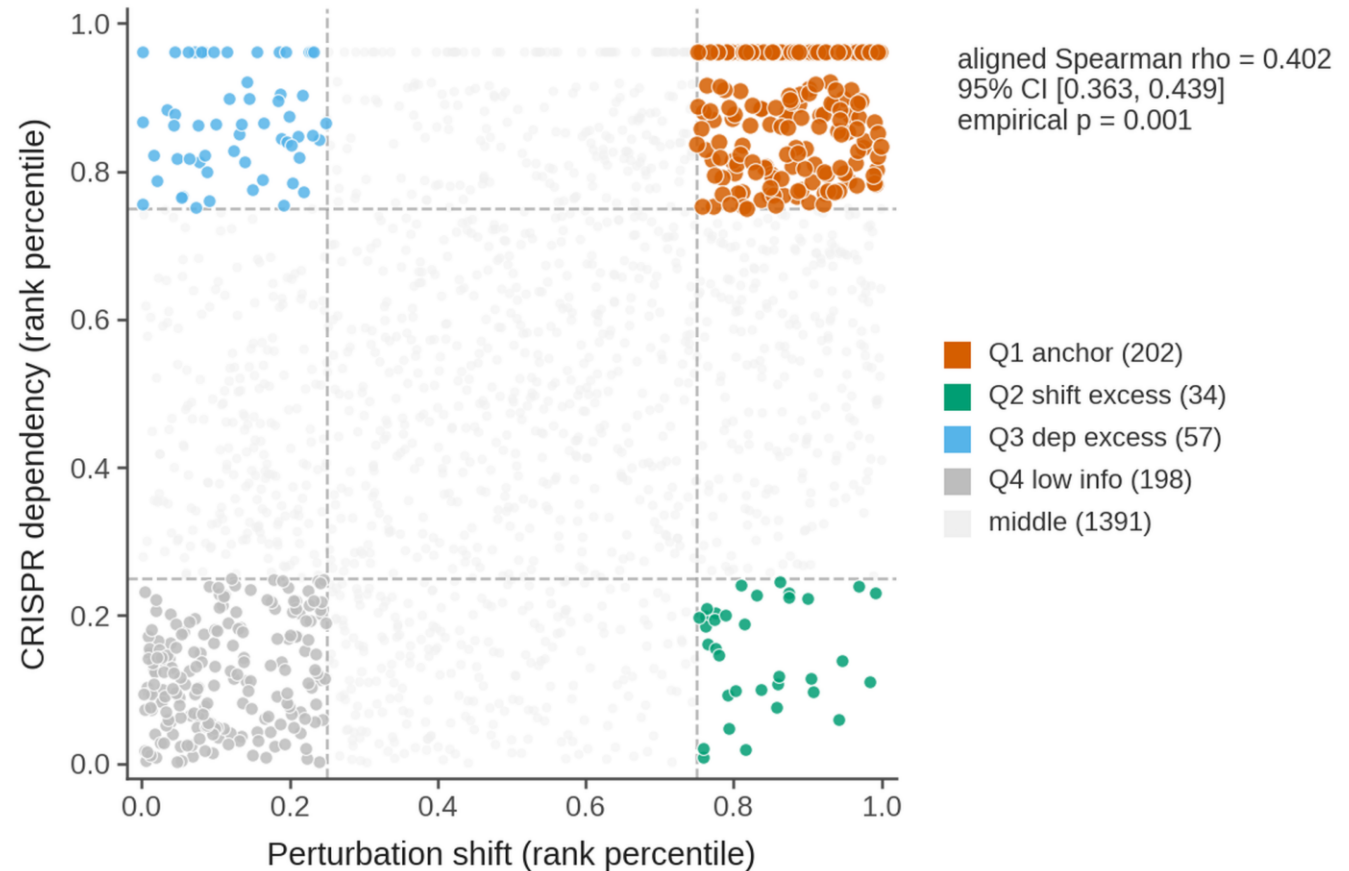
Extended Data Fig. 2 | Robustness of perturbation-fitness bridge metric selection

a, Top-n gene-subset sensitivity. b, Metric and CRISPR endpoint robustness. The heat map shows mean aligned Spearman correlation across HCC38 and HCC1143 for candidate truth metrics crossed with the two CRISPR DepMap readouts; the boxed cell marks the retained primary metric and endpoint. c, Endpoint sensitivity. d, Control-subsampling robustness. Aligned Spearman correlation with DepMap CRISPR dependency was recomputed across repeated control-cell subsamples for candidate truth metrics in HCC38 and HCC1143; points show the mean and horizontal intervals show the 2.5th-97.5th percentile range across subsamples. e, Whole-transcriptome shift versus target-gene self-expression in driving the fitness bridge. Each point represents one perturbation target in HCC38 (left sub-panel) or HCC1143 (right sub-panel). The x-axis shows the target gene's own log2 fold-change; the y-axis shows DepMap CRISPR dependency.

a K562 temporal stratification (7d vs 13d)

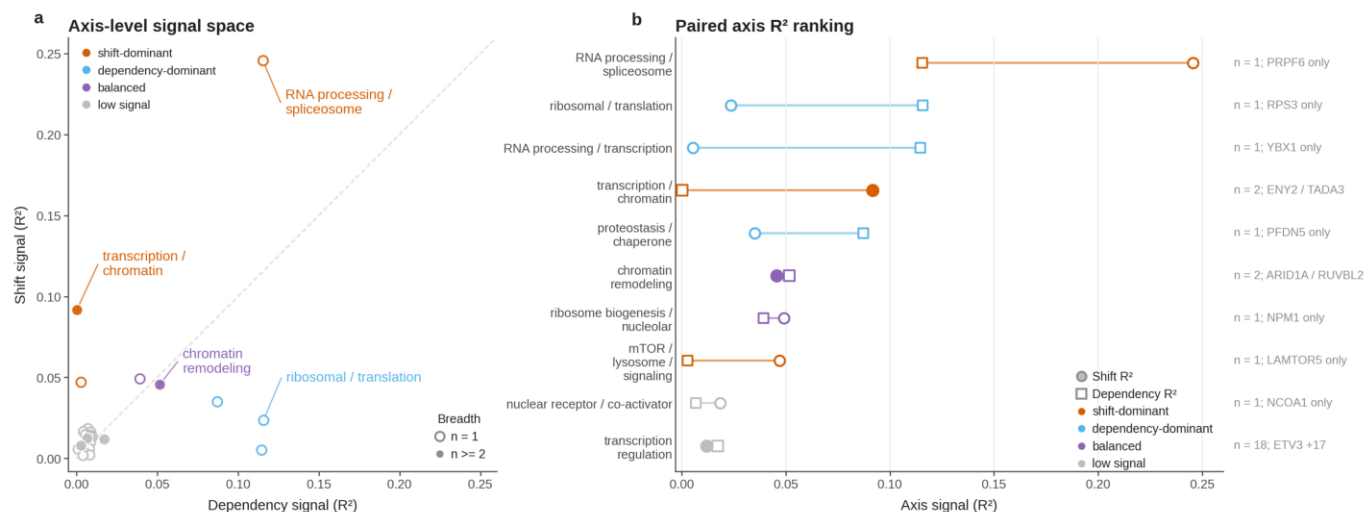


b Large-scale perturbation-fitness bridge confirmation



Extended Data Fig. 3 | K562 temporal evidence and large-scale bridge confirmation

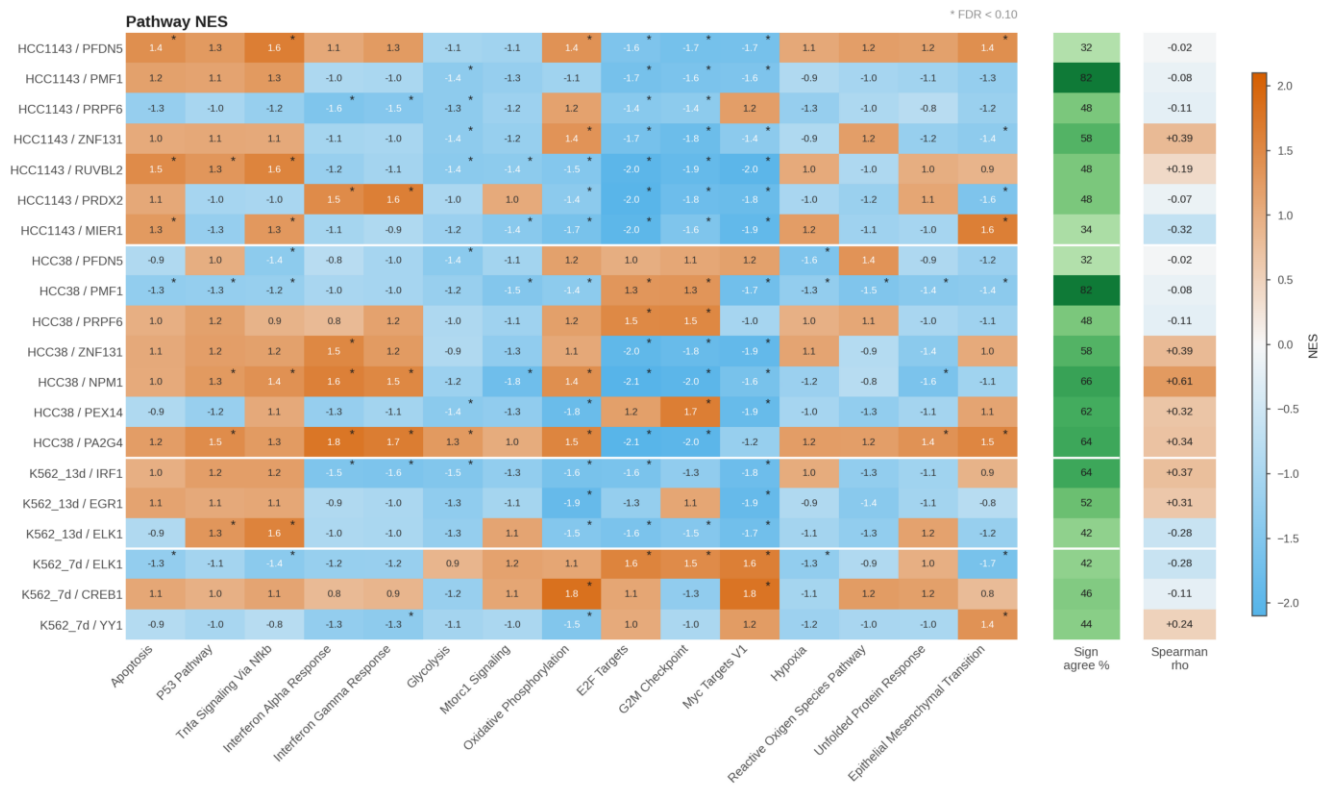
a, K562 temporal bridge-magnitude dissociation (GSE90063). Aligned Spearman correlation with CRISPR DepMap dependency and mean perturbation-shift magnitude are shown for the 7d temporal-sensitivity panel and the 13d formal supplementary bridge test. b, Replogle K562 essential day 7 large-scale perturbation-fitness bridge. Scatter of `real_shift_mean_abs` versus `depmap_gene_dependency` for 1,882 matched CRISPRi targets, displayed as within-context rank percentiles (0–1 axes, matching the convention of Fig. 1c,d). Dashed lines mark the 25th and 75th percentiles.



Extended Data Fig. 4 | Descriptive axis-level signal space

a, Axis-level signal space comparing dependency signal with transcriptomic shift signal across annotated axes. b, Paired axis R-squared ranking. Axes with the largest value in either signal dimension are displayed with matched shift and dependency R-squared values on the same scale; labels report compact target breadth.

a Exploratory pathway-level summaries of target-control perturbation responses



Extended Data Fig. 5 | Exploratory pathway-response polarity heatmap

a, Pathway normalized enrichment score (NES) heat map across the pre-specified display targets in HCC38, HCC1143, K562 7d, and K562 13d. Rows show selected benchmark anchors and high-variance response examples; columns show the pre-specified Hallmark response panel. Asterisks indicate pathway enrichments passing FDR < 0.10. Right-hand summaries report same-target partner-context pathway Spearman correlation and sign-agreement fraction across all Hallmark pathways.